

SPPH 604 — Project Increments P1–P5

These five weekly increments turn your **M0-locked paper** into your **M1 replication**. Each week you apply that week’s lab skill to *your own* paper and commit one small, reproducible piece to your group repo. By the end of P5 you have assembled M1 — and you have practised the two things assessed after it: critiquing another group’s M1 at M2, and defending your own analysis, alone, at M3.

The one question you are answering all term

You keep **one scientific question** fixed and interrogate it from five angles. For the demonstration paper it is:

Among adults meeting the study’s MASLD criteria, is central adiposity associated with all-cause mortality — and how robust is that to cohort construction, confounding, effect heterogeneity, survey design, and missing data?

Write the equivalent one-sentence question for **your** paper in P1 and keep it at the top of every increment.

Every increment runs the same four moves

1. **Replicate** — what did the authors do, and can you reproduce it?
2. **Interrogate** — which analytic choice matters most here?
3. **Improve** — implement one defensible alternative.
4. **Defend** — does it change the scientific interpretation, and why?

The five increments

Note the **+1 offset**: L1 (shortlist) has no increment, so each P lines up with the *next* lab number.

#	Increment	Week	Applies lab	Feeds M1 slide
P1	Analytic cohort	3	L2 Analytic data	Cohort flow diagram
P2	Confounding diagnostic (Table 1)	4	L3 Confounding	Table 1 + variable roles
P3	Effect modification (interaction)	5	L4 Interaction	Effect-modification slide
P4	Design-aware main estimate	6	L5 Survey analysis	Reproduced Table 2 / KM
P5	Missing-data sensitivity	7	L6 Missing data	Missingness audit

These increments are **optional and ungraded** — there is no separate weight, no separate deadline, and nothing to submit. Keep each one in your **group GitHub repo** as you go, because **M1 is assembled from these five increments**: doing them week by week is what stops M1 from becoming five weeks of work in one. Each handout ends with a *What good looks like* section — use it to check your own work — and a **Be ready to defend** question, which is the kind of thing you will be asked live at M3.

Where “the dominant threat” went: into M1

Naming the **single most consequential threat** to the paper’s conclusion is a *synthesis* judgment — you cannot rank threats before you have actually done the confounding (P2), interaction (P3), design (P4), and missingness (P5) work. So it is **not a weekly increment**; it is the analytic-judgment core of **M1**. At M1 you:

1. present your P1–P5 replication, then
2. argue which one threat could most change the interpretation (for the demo paper, a strong candidate is *exposure/selection circularity* — the phenotype groups and the FLI eligibility filter both use BMI and waist), and
3. run or propose one targeted check on it.

That “biggest threat” argument is what your **M2 reviewers will interrogate**, and what you defend at M3.

Ground rules for every increment

- **Do not force the paper’s exact numbers.** A discrepancy you have *traced and explained* beats an unexplained exact match, or code that hard-codes the target.
- **Justify, don’t automate, your choices.** “Adjust for everything,” AIC/stepwise confounder selection, and hard-coded imputation settings are lab conveniences, not project methods. Argue from causal/temporal reasoning.
- **AI co-pilots are allowed — and you are accountable.** Each handout names a mistake a co-pilot can make at that step. You must find and fix it, and you must be able to answer that increment’s **defence question** live at M3, alone.

Optional: a one-page note on what you decided

Nothing in this section is collected, read, or marked. It is a habit some groups find useful, offered here because it costs about a page and it makes five separate increments easier to assemble into one M1.

Once your paper is locked, the seven things below are already settled somewhere in your group’s head — establishing them is what M0 asked of you. Writing them down in one place, early, gives you a record of what you believed *before* you had seen any of your own results:

1. **The scientific question** — one sentence, in the terms your own analysis will use.
2. **The primary contrast and estimand** — which two groups are compared, on what scale, and in which population.
3. **The exposure and outcome definitions** — by variable name and cut point, not by description.
4. **The prespecified adjustment set** — the causal or temporal reason for each variable you include, and for any important one you deliberately leave out.
5. **The effect modifier you intend to examine** — and why that one rather than another.
6. **The design features your analysis has to respect** — named, with the variables that encode them (weights, strata, and clusters, where your data have them).
7. **The missingness you expect to have to handle** — which variable of your locked analysis carries it, and roughly how much.

Where you keep it is up to you: a file in your group repo, a shared document, or a comment block at the top of your P1 script. There is no template to fill in, no place to hand it in, and no format anyone is checking.

The point is not to be right the first time. You will change your mind about several of these, and changing your mind for a stated reason is good epidemiology. The point is that a decision you wrote down in advance is one you can notice yourself revising — which is exactly the kind of reasoning you present at M1 and defend at M3.

The demonstration paper

Each handout contains a *Worked example* box from the instructor’s full reproduction of **Kueh et al., BMJ Open 2026;16:e113719** (MASLD/NHANES). Use it to see the target and the traps — not to copy. Your paper is different; the *moves* are the same.

Two analyses of the same cohort

The worked examples in these handouts are a **paper replication**: they keep the paper's four phenotype groups (I–IV) and its Model 1 and Model 2 covariates, and report the paper's contrast (Group IV vs I) beside our locked one (IV vs III). The Week 4–7 lectures run a second analysis of the same cohort, an **exposure analysis** of obesity and of high waist fat within obesity strata, adjusted only for factors that precede the exposure (age, sex, race/ethnicity, current smoking and sedentary time), because Week 4 argues that several Model 1 covariates may lie on the causal path. The two can give different numbers for what looks like the same comparison; the rows differ only slightly, so most of the difference comes from the adjustment set. Results in these handouts are therefore labelled with the comparison, the rows used, the adjustment set and whether the estimate is survey-weighted.